

Intel APX and x86S Modernization

<https://gemini.google.com/app/cafbe2119feb04c4>

User prompt: Review: <https://www.phoronix.com/news/x86-CPU-With-APX-x86S>

Response: Phoronix's coverage of **Intel's APX (Advanced Performance Extensions)** and **x86S (64-bit Simplified Architecture)** highlights one of the most fundamental modernization efforts for the x86-64 architecture since its creation.

Overview: The Twin Pillars of x86 Modernization

Intel's combined push for **APX** and **x86S** addresses the two biggest critiques of the x86 architecture: legacy bloat and register scarcity relative to ARM64/AArch64 and RISC-V.

Initiative	Primary Goal	Key Architectural Changes
Intel APX	Boost performance & instruction efficiency	Doubles GPRs (16 → 32), introduces 3-operand format, flag suppression
Intel x86S	Streamline hardware & remove legacy debt	Eliminates 16-bit real/protected mode, 32-bit ring 0/1/2/3 legacy modes

1. Intel APX (Advanced Performance Extensions)

APX is a major set of instruction set architecture (ISA) extensions designed to increase execution efficiency without requiring radical changes to silicon thermal or power budgets.

- **32 General-Purpose Registers (GPRs):** Expands the GPR set from 16 (R0–R15) to 32 (R0–R31). This reduces register spilling and memory stack pushes/pops by up to 10–20%, bringing x86 GPR count to parity with ARM64 and RISC-V.
- **3-Operand Syntax (NDD - Non-Destructive Destination):** Legacy x86 generally overwrites one of its inputs (e.g., `ADD RAX, RBX` overwrites `RAX`). APX introduces 3-operand forms (e.g., `ADD RDX, RAX, RBX`), preserving source registers without extra `MOV` instructions.
- **Conditional Operations (CCMP, CTEST):** Adds conditional compare and test instructions to allow complex logical evaluation without branching, reducing branch mispredictions in critical loops.
- **Status Flag Suppression:** Allows instructions to opt out of updating CPU status flags (e.g., `ZF`, `CF`), preventing unnecessary register dependency chains in out-of-order execution pipelines.
- **Optimized Memory Operations:** Introduces multi-register push/pop operations (`PUSH2` / `POP2`) for faster function prologues and epilogues.

2. Intel x86S (Simplified 64-bit Architecture)

x86S is a specification proposed by Intel to eliminate legacy operating modes from hardware silicon altogether.

- **Pure 64-Bit Booting:** Eliminates 16-bit real mode and 16-bit/32-bit protected mode hardware startup pathways. The CPU powers up directly into 64-bit mode (Long Mode), relying strictly on 64-bit firmware (UEFI).
- **Removal of Legacy Artifacts:** Strips out support for legacy ring 1 and ring 2 privilege levels, 16-bit addressing, legacy 32-bit paging structures, 8259 PIC controllers, and the A20 gate mechanism.
- **Simplified Silicon Design:** Reducing architectural legacy modes cuts down the validation matrix, shrinks the CPU attack surface, and frees up die area for functional microarchitectural improvements.
- **32-Bit Compatibility Strategy:** While ring 0 through ring 3 native hardware execution for 32-bit is stripped, 32-bit legacy user space applications are intended to run via light user-space translation or virtualized environments.

Ecosystem & Toolchain Status

As detailed across Phoronix's reporting, open-source toolchains and developer tooling have actively integrated early support for these capabilities:

- **Compiler Support:** GCC, Clang/LLVM, and GNU Binutils received patches adding assembler support, register allocation tweaks, and `-mapx` compilation flags.
- **Emulation & Testing:** Intel enabled APX and x86S within the Intel Software Development Emulator (SDE) and QEMU, allowing developers to build and benchmark APX-optimized binaries prior to physical silicon deployment.
- **Kernel Integration:** Linux kernel developers began restructuring boot procedures to accommodate 64-bit-only boot environments and handling the expanded context switches required for 32 GPRs.

Key Takeaways

1. **Cleaner Code, Better IPC:** APX delivers a measurable bump in instructions-per-clock (IPC) and code density by eliminating redundant MOV operations and stack spills.
2. **Modern Boot Stack:** x86S officially sheds 40+ years of backward-compatibility baggage, modernizing x86 hardware closer to contemporary pure 64-bit ISA design standards.
3. **Competitive Parity:** Together, APX and x86S reposition x86-64 against ARM64 and RISC-V in data center and client compute efficiency.

User prompt: "The exact Atom technology Rosaic will receive remains unclear, although the startup could potentially gain access to Tremont, one of Intel's final Atom microarchitectures. Tremont was used in products such as Elkhart Lake and Jasper Lake, which supported embedded systems, low-power laptops, and Chromebooks." <https://www.techradar.com/pro/why-is-intel-sharing-its-atom-cpu-technology-with-a-start-up-the-answer-could-be-in-low-power-x86-design> <https://www.techpowerup.com/cpu-specs/atom-x6200fe.c4283> The die space says 64mm². Can this processor be modified to run at 0.4V near threshold voltage and 100MHz using an X86S ISA and run a single core in less than 4mm²?

Response: **Yes, conceptually and architecturally, a modified Tremont core could easily fit into less than 4 mm² and run at 0.4V at 100 MHz using x86S.** However, this would require re-synthesizing and laying out the physical design with low-voltage cell libraries rather than just under-volting standard off-the-shelf silicon.

Here is a technical breakdown of how the area, voltage/frequency scaling, and ISA modifications interact:

1. Core Area Analysis (< 4 mm²)

The 64 mm² die size of the **Atom x6200FE** (Elkhart Lake platform) includes not just the CPU cores, but the entire System-on-Chip (SoC):

- Integrated Intel Gen11 Graphics (GPU)
- Memory Controllers (LPDDR4/DDR4)
- Integrated PCH / I/O interfaces (PCIe, USB, Ethernet, GPIO)
- System Agent & Uncore interconnects

Tremont Native Core Size

On Intel's 10nm (SuperFin) process node:

- A complete **4-core Tremont module** (including a shared 1.5 MB L2 cache and interconnects) occupies roughly **4.5 mm² to 5.5 mm²**.
- A **single Tremont CPU core** (execution engine + 32KB I-Cache + 32KB D-Cache) occupies only **~0.6 mm² to 0.9 mm²**.

Even with private L2 cache allocated to a standalone core, a single Tremont core occupies **well under 2 mm²**, easily meeting your **< 4 mm²** target.

2. Voltage and Frequency Scaling (0.4V @ 100 MHz)

Running a CPU at 0.4V enters the **Near-Threshold Voltage (NTC)** computing regime, where the supply voltage (V_{dd}) approaches the transistor threshold voltage ($V_{th} \approx 0.3V - 0.35V$).

Frequency Alignment

- At nominal voltage (~0.9V–1.1V), Tremont runs up to 3.0 GHz.
- When dropping V_{dd} to 0.4V, transistor drive current drops exponentially, increasing gate delays by 10× to 30×.
- Scaling frequency down from **~3.0 GHz to 100 MHz** (a 30× reduction) perfectly matches this gate delay curve.

Hardware Modification Challenges

While 100 MHz at 0.4V is physically achievable, stock Tremont silicon cannot simply be under-volted to 0.4V without modification:

1. **SRAM V_{min} Wall:** Standard 6T SRAM cells used in L1/L2 caches generally experience read/write instability below ~0.55V–0.6V. To operate at 0.4V, the caches must be re-architected using **8T/10T SRAM cells** or low-voltage assist circuits.
2. **Timing Variations:** At 0.4V, process and temperature variations impact signal propagation much more severely. The pipeline would need to be synthesized using **Wide-Voltage-Range (WVR)** or ultra-low-power standard cell libraries.

3. Impact of the x86S ISA

Adopting **x86S** (Intel’s 64-bit Simplified Architecture) directly aids low-voltage, small-area implementation:

- **Removal of Legacy Decoders:** x86S eliminates hardware support for 16-bit real mode, 16-bit/32-bit protected mode, ring 1/2 privileges, and legacy segment limit checks.
- **Reduced Front-End Area:** Legacy decode logic in x86 is complex and power-dense. Stripping it simplifies the instruction fetch and decode pipeline, shrinking front-end die area by a few percent and reducing switching noise.
- **Simplified Timing Paths:** Fewer legacy state transitions reduce critical path complexity, making static timing analysis (STA) and timing closure significantly easier at near-threshold voltages.

Summary

Metric	Stock Tremont (10nm)	Target Modified x86S Core	Feasibility
Single Core Area	~0.8 mm ² (Core + L1)	< 4.0 mm ²	Easily Achieved (Leaves ~3 mm ² for extra L2 or custom IP)
Operating Voltage	~0.8V – 1.1V	0.4V (Near-Threshold)	Feasible , requires 8T/10T SRAM & WVR cell synthesis
Clock Frequency	1.5 GHz – 3.0 GHz	100 MHz	Optimal match for 0.4V gate delays
ISA Architecture	x86-64 (Full Legacy)	x86S (Pure 64-bit)	Ideal (Simplifies decoders and lowers power/area)

A single-core, x86S-based Tremont derivative re-synthesized for low voltage would represent a tiny (~1 mm²), sub-milliwatt x86 engine capable of running standard 64-bit Linux code at 100 MHz.

User prompt: Can IGZO 0C2T eDRAM be synthesized with this modified X86S core at 0.55V?

Response: **Yes, absolutely.** Integrating IGZO 2T0C (0-Capacitor, 2-Transistor) eDRAM with a 0.55V modified x86S core is not only technically feasible, but architecturally an ideal match for ultra-low-power, compact computing.

1. Monolithic 3D BEOL Integration (Area & Process)

- **Zero Front-End Silicon Footprint:** IGZO thin-film transistors (TFTs) can be processed at low temperatures (< 300°C), allowing them to be fabricated during Back-End-of-Line (BEOL) metal interconnect layer processing.
- **Area Savings:** Because the IGZO eDRAM cache macro sits directly *above* the silicon CMOS core in 3D, it adds zero additional Front-End-of-Line (FEOL) silicon area. This ensures the x86S core + cache easily stays well under your **4 mm²** target (with the core logic itself occupying ~ 0.5 mm²–1.0 mm²).

2. Voltage (0.55V) & Timing (100 MHz) Alignment

- **SRAM V_{min} Bypass:** Standard 6T SRAM caches struggle at 0.4V–0.55V due to degraded Static Noise Margin (SNM) and read-disturb issues. IGZO 2T0C eDRAM avoids this because read and write paths are physically decoupled across two separate transistors.
- **Frequency Match:** At 100 MHz, a single clock cycle is 10 ns. Advanced IGZO 2T0C cells achieve write and read latencies under 10 ns, allowing the cache to operate at full speed without inserting wait states.
- **Near-Zero Refresh Overhead:** IGZO has an extremely wide bandgap, yielding ultra-low off-state leakage ($I_{off} < 10^{-19}$ A/\μm). This extends charge retention times to over 100 seconds (compared to ~ 64 ms for standard DRAM), effectively turning the eDRAM into a quasi-non-volatile cache that consumes virtually no static refresh power at 0.55V.

3. Design Implementation & Synthesis Nuance

- **Macro Compilation vs. RTL Synthesis:** While the x86S CPU logic (decoders, execution units, registers) is synthesized from RTL using standard logic cell libraries, the IGZO eDRAM array itself is instantiated as a compiled memory macro (using specialized BEOL memory layout compilers).
- **Peripheral Voltage Drivers:** While the storage cells and read circuits operate natively at 0.55V, the BEOL wordline drivers can utilize local charge pumps or back-gate biasing to slightly boost write voltages (V_{WL}) internally to guarantee fast charge injection into the storage node without raising the global system V_{dd} .

Summary

Metric	Standard Silicon SRAM Cache	IGZO 2T0C BEOL eDRAM
Placement	FEOL Silicon (Takes die space)	BEOL Interconnect (Stacked on top)
0.55V Stability	High bit-flip risk (requires 8T/10T cells)	Excellent (separated read/write paths)
Standby Leakage	High subthreshold leakage at low V_{th}	Near-zero ($I_{off} < 10^{-19}$ A/ μ m)
100 MHz Latency	< 2 ns	< 10 ns (Single-cycle at 100 MHz)

Pairing a synthesized 0.55V x86S core with a monolithically stacked 3D IGZO 2T0C eDRAM cache creates a sub-milliwatt x86 engine with extreme density and near-zero standby power.

User prompt: how much space would be required to fit 64MB of EDRAM using this process?

Response: Fitting **64 MB (512 Mb)** of IGZO 2T0C eDRAM requires between **0.6 mm² and 2.4 mm²** of total macro area using modern Back-End-of-Line (BEOL) processes, depending on the minimum feature size (F) and cell stacking architecture.

Footprint Estimates Across BEOL Process Nodes

BEOL Node (F)	Cell Architecture	Cell Footprint	Raw Array Area	Total Macro Area (~70% Efficiency)	With 2-Bit MLC (Multi-Level Cell)
14 nm	3D-Stacked ($4F^2$)	$0.00078 \mu\text{m}^2$	0.42 mm^2	$\sim 0.60 \text{ mm}^2$	$\sim 0.30 \text{ mm}^2$
20 nm	3D-Stacked ($4F^2$)	$0.00160 \mu\text{m}^2$	0.86 mm^2	$\sim 1.23 \text{ mm}^2$	$\sim 0.61 \text{ mm}^2$
28 nm	3D-Stacked ($4F^2$)	$0.00314 \mu\text{m}^2$	1.68 mm^2	$\sim 2.41 \text{ mm}^2$	$\sim 1.20 \text{ mm}^2$
40 nm	3D-Stacked ($4F^2$)	$0.00640 \mu\text{m}^2$	3.44 mm^2	$\sim 4.91 \text{ mm}^2$	$\sim 2.45 \text{ mm}^2$
28 nm	Planar 2T0C ($16F^2$)	$0.01254 \mu\text{m}^2$	6.73 mm^2	$\sim 9.62 \text{ mm}^2$	$\sim 4.81 \text{ mm}^2$

Note: Calculations assume binary 64 MB = 536.87 Mbit ($64 \times 2^{20} \times 8$ bits). Macro area includes a 30% area overhead for sense amplifiers, wordline drivers, and local decoders.

Architectural Scaling Factors

- Vertical 3D Stacking ($4F^2$ Footprint):** Stacking the write transistor (T_W) in a upper BEOL layer directly over the read transistor (T_R) reduces the horizontal cell footprint from a planar $16F^2$ – $20F^2$ down to $4F^2$, matching standard vertical DRAM cell density.
- Multi-Bit Storage (MLC Advantage):** Because IGZO's off-state leakage is virtually zero ($I_{off} < 10^{-19}$ A/ μ m), the charge stored on the read-transistor gate node remains stable over long periods (> 100 seconds). This enables reliable 2-bit (4-voltage level) storage per cell, cutting the required physical cell count—and physical area—in half.
- CMOS-Under-Array (CUA) Integration:** Because the IGZO memory array resides entirely within the BEOL metal layers above the silicon substrate, peripheral logic (sense amplifiers, control logic, address decoders) can be placed in the underlying Front-End-of-Line (FEOL) silicon. If placed directly underneath the memory macro or integrated into the adjacent x86S core area, zero additional silicon substrate area is wasted.

In a 20nm 3D BEOL process, a 64 MB IGZO eDRAM cache macro sitting directly atop the $\sim 1 \text{ mm}^2$ modified x86S core occupies approximately **1.23 mm²** of BEOL area, leaving the combined CPU + Cache system well under the **4 mm²** target.